EU Law and the Energy Transition: Leaving Soils Behind?

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Abstract Despite providing essential ecosystem functions and being vital for human well-being, soil remains the most underregulated natural component within European Union (EU) law, lacking an overarching legal framework. While this omission has been previously identified, this chapter aims to provide a comprehensive overview of how the EU has balanced the pressing needs arising from energy transition policy on the one hand, and the EU soil policy goals of protecting, restoring, and sustainably managing soil on the other, as prescribed by the integration principle. To this end, we begin by offering a critical overview of EU climate and energy laws and policies. In order to provide a comprehensive analysis of how soil impacts are considered, we also examine other laws and policies that shape the broader context of the energy transition, including regulations on critical raw materials and sustainable investment. Finally, we analyse how the EU Nature Restoration Law addresses soil scarcity in the context of pressing energy policies. We conclude that the difficult choice between protecting soil ecosystems and advancing the energy transition has been resolved in EU law in favour of the latter.

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1 Introduction

It is a platitude to state that soil provides crucial services for ecosystems and human well-being.[1] Nevertheless, soil remains the most underregulated of all natural components in the European Union (EU).[2] The EU Soil Strategy for 2030[3] reiterates the need to adopt protection, restoration, and sustainable use measures to safeguard soil's ecological functions, particularly for the benefit of biodiversity, food security, climate change mitigation, and economic development. Consequently, soil management measures are structurally intertwined with other regulatory and policy fields, including those relating to pollution control, biodiversity conservation, agriculture and climate change. As a result, overlapping policy objectives—such as renewable energy production, carbon offsetting, climate change adaptation, restoration of habitats, or food security—give rise to a multitude of competing, and often conflicting, land uses. The role of healthy soils in mitigating climate change is commonly pinned to carbon storage, leading to the assumption that climate and soil policies create a positive synergy.[4] While this is true, a comprehensive assessment leads to a more complex picture, including direct and indirect negative interactions, particularly in what concerns one of the most critical EU priorities: the energy transition.[5]

Delivering a climate-neutral economy appears at the top of the environmental agenda, as designed in the European Green Deal,[6] requiring the production and distribution of clean energy, and a significant set of legal acts has been adopted to facilitate this transition. Notably, as a pillar of the EU's energy policy, renewable

[1] An overview is provided in FAO and ITPS (2015, p. 13).

[2] As summarised by Heuser (2022, p. 2) *"soil has been subject to several sector-related policy measures in the Union, but a special soil protection policy is still in its infancy (…). The current legislation covers some aspects of soil pollution and biological soil protection, but it remains fragmented and achieves soil protection only as side effects. Furthermore, EU law hardly considers the requirements to prevent soil compaction, erosion, sealing and other physical threats of the soil."*

[3] Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions "EU Soil Strategy For 2030 reaping the benefits of healthy soils for people, food, nature and climate" (COM/2021/699 final).

[4] See, among others, Hannam (2024a).

[5] A significant amount of research has been produced in the field, with a focus on impacts on soil, water and ecosystems—cf., among many others, Hastik et al. (2015), Lal (2009), Rahman et al. (2022), Sayed et al. (2021), Sokka et al. (2016).

[6] The European Green Deal was adopted by the European Commission as a strategic roadmap for dealing with environmental challenges, including climate change, with the goal of delivering a sustainable economy and achieving climate neutrality by 2050. It outlines an integrated approach to environmental, economic, and social transformation, covering sectors such as energy, industry, agriculture, and biodiversity. Key objectives include reducing greenhouse gas emissions by at least 55% by 2030 (compared to 1990 levels), decoupling economic growth from resource use, and ensuring no one is left behind in the transition. The Green Deal serves as the overarching framework for legislation such as the European Climate Law, the Fit for 55 package, and the Nature Restoration Law. Cf. the Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions, The European Green Deal, COM (2019) 640 final.

energy generation should be driven by the goal of preserving and improving the environment.[7] This aligns with the general principle of sustainable development as enshrined in Article 3(3) Treaty of the European Union (TEU), as well as the principle of Environmental Integration pursuant to Article 11 Treaty on the Functioning of the European Union (TFEU).[8] Together, these principles support the EU's broader objective of achieving a high level of protection and improvement of the environment within and beyond its borders according to Article 191(2) TFEU and Article 21(2) TEU.[9]

Furthermore, and importantly, the legal framework governing soils in the EU remains utterly fragmented.[10] While a proposal for uniform EU legislation (the 'Soil Monitoring Law') is currently under scrutiny,[11] provisions related to soil protection and management are enshrined across different pieces of EU law dealing, among others, with energy and environmental matters. EU law provisions applicable to soils can thus be found across legally binding legislation such as Regulations (which are directly applicable in all Member States) and Directives (which instead require transposition by Member States in their domestic legal frameworks), as well as soft law documents, such as the Soil Strategy for 2030 and the EU Biodiversity Strategy for 2030.[12] An analysis of the legal framework for soil protection under EU law must therefore rely mostly on how these acts interplay with energy laws and policies in the context of the energy transition.

Against this backdrop, this chapter seeks to offer a comprehensive analysis of how the EU has navigated the tension between the urgent demands of energy transition policies and its soil policy objectives—namely, the protection, restoration, and sustainable management of soil—as required by the above-mentioned Environmental Integration principle. To this end, we provide a review of how EU climate and energy binding and non-binding normative acts have considered potential impacts on soil health (Sect. 2). However, fully understanding this multifaceted challenge requires analysis beyond energy law. Energy transition depends on other factors and policies, such as those concerning critical raw materials and sustainable investment (Sect. 3). Moreover, the legal framework created for energy transition is only one of the sides in what ought to be integrated environmental, climate, and energy policies. It is also crucial to understand how recent EU laws and proposals with a focus on ecological conservation, including soil, address soil scarcity in the context of

[7] Article 194(1) TFEU.

[8] Cf., Bándi (2020, pp. 38–43); see also Opinion of Advocate General Jacobs, Case C-379/98, *Preussen Electra*, ECLI:EU:C:2000:585, para 231.

[9] For an overview of the procedural and substantive framework surrounding the EU's external environmental competence, cf. Vedder (2012, p. 13).

[10] See Paleari (2017) and Heuser (2022).

[11] Proposal for a Directive of the European Parliament and of the Council on Soil Monitoring and Resilience, (COM/2023/416 final).

[12] EU Soil Strategy for 2030; Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions, "EU Biodiversity Strategy for 2030: Bringing nature back into our lives" (COM/2020/380 final).

pressing energy policies. With this goal in mind, we will turn our attention to nature restoration (Sect. 4).

2 Soils in EU Climate and Energy Law

This section analyses key legislation governing the broader EU Climate and Energy Framework, with a specific focus on EU legislation on renewable energy generation. It aims to unfold and underscore the role of soil management and protection under both the overarching EU's climate governance framework and in the related planning requirements for Member States, as well as in relation to pointed EU law provisions and approaches to soil protection with regard to the production of biofuels and biomass, as well as the deployment of renewable energy installations.

2.1 EU Climate Law and Governance

As mentioned above, soils are key to climate change mitigation as their degradation leads to a decrease in natural carbon sinks. On the flipside, healthy soils store large quantities of carbon (C) in the form of soil organic carbon (SOC). In this connection, soil management plays a crucial role in enhancing the carbon cycle and maintaining the global carbon balance. It therefore provides multiple benefits in terms of climate change mitigation and adaptation, thus actively contributing to multiple goals set out in the EU's climate policy. More specifically, looking at the EU's Soil Strategy for 2030, not legally binding targets are set in terms of achieving land-based climate neutrality in the EU by 2035, as a conducive step towards the legally binding climate neutrality goal by 2050 enshrined in the EU Climate Law.[13] Moreover, as part of the comprehensive definition of soil health embraced in the strategy by the EU, the soil functions as a carbon reservoir is highlighted alongside other functions such as biodiversity conservation and protection of water bodies.

The EU Climate Law is the overarching piece of legislation dealing with climate action in the EU since it entered into force in 2021.[14] The goal of the EU Climate Law is to establish *"a framework for the irreversible and gradual reduction of anthropogenic greenhouse gas emissions by sources and enhancement of removals by sinks"* consistent with the global temperature goal set out under Article 2(1) of

[13] Regulation (EU) 2021/1119 of the European Parliament and of the Council of 30 June 2021 establishing the framework for achieving climate neutrality and amending Regulations (EC) No. 401/2009 and (EU) 2018/1999.

[14] Regulation (EU) 2021/1119 of the European Parliament and of the Council of 30 June 2021 establishing the framework for achieving climate neutrality and amending Regulations (EC) No. 401/2009 and (EU) 2018/1999.

the Paris Agreement.[15] While soil is not directly addressed in the European Climate Law, it is somehow embedded as one of the environmental components of the forest ecosystem as a key carbon sink for the purposes of greenhouse gas (GHG) emissions mitigation. The European Climate Law emphasises the role of terrestrial ecosystem restorations in terms of enhancing natural carbon sinks, with a specific focus on forests as carbon sinks, storage and substitution.[16] Therefore, a revision of the Land Use, Land Use Change and Forestry (LULUCF) Regulation[17] as part of the 'Fit-for-55' package[18] was launched by the European Commission to implement the targets set out in the European Climate Law.[19] These developments should moreover be read and understood against the background of the EU Soil Strategy for 2030. The Strategy foregrounds the sustainable use and management of soils as a key solution to climate mitigation in the EU and embraces the goal of 310 million tCO_2 by 2030 from the land-use sector.[20] Moreover, it includes carbon sequestration of soils as one of the tenets of soil health, with particular focus on organic soils and mineral soils.[21] Protection and restoration of peatlands and wetlands in the EU, as well as the enactment of measures to ensure ecological integrity and restoration of soil ecosystems with high SOC, are seen as a major mitigation option, as backed by supporting carbon farming and low-carbon agricultural practices consistent with the Farm-to-Fork strategy and the Biodiversity Strategy for 2030.[22]

The European Climate Law establishes a mid-term climate mitigation target of −55% GHG emissions compared to 1990 levels. Within such a target, it specifies that the contribution of net removals to the Union 2030 climate target shall be limited to 225 million tonnes of CO_2 equivalent, with a higher volume of carbon sinks to be adopted after 2030.[23] Another mid-term target for 2040 under discussion at the time of writing, which builds upon the proposal submitted by the European Scientific Advisory Board on Climate Change, which might include up to 400 $MtCO_2e$ of

[15] Article 1 European Climate Law.

[16] Id., Recitals 22 and 23.

[17] Regulation (EU) 2018/841 of the European Parliament and of the Council of 30 May 2018 on the inclusion of greenhouse gas emissions and removals from land use, land use change and forestry in the 2030 climate and energy framework, and amending Regulation (EU) No. 525/2013 and Decision No. 529/2013/EU.

[18] Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions "Fit for 55: delivering the EU's 2030 Climate Target on the way to climate neutrality," COM/2021/550 final.

[19] On this development, see Heuser (2022, pp. 3–4).

[20] Cf. Section 2, under medium-term objectives by 2030 – EU Soil Strategy For 2030, p. 3.

[21] EU Soil Strategy, pp. 6–7.

[22] Ibid., see also Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions, "A Farm to Fork Strategy for a fair, healthy and environmentally-friendly food system" (COM/2020/381 final); see also EU Biodiversity Strategy for 2030.

[23] Article 4(1) European Climate Law.

carbon removals, including removals by sinks.[24] Lastly, the EU Climate Law enshrines a climate neutrality goal by 2050, which should largely comprise the enhancement of natural carbon sinks (foremost forests) and the deployment of carbon removal techniques, including carbon farming.[25] Importantly, the targets set out in the European Climate Law are legally binding for the EU and its Member States and entail far-reaching procedural obligations.[26] This, in turn, should imply, at least in principle, a requirement upon EU Member States as to the measures to be adopted to restore and protect soils as natural carbon sinks to achieve the European Climate Law targets.[27]

The implementation of the mid- and long-term climate targets is left to a complex mechanism of reporting, which lies foremost in the hands of Member States. Accordingly, most of the concrete measures adopted by Member States should be outlined by Member States in their National Energy and Climate Plans (NECPs) drafted under the EU Governance Regulation.[28] The submitted NECPs and the domestic measures contained therein are iteratively scrutinised and assessed by the European Commission to ensure consistency, among others, with the European Climate Law and the EU's renewable energy targets.[29]

However, a recent account of the submitted NECPs shows both little focus on soil monitoring and wide heterogeneity across Member States as to their approach to soil management, protection, and restoration. [30] For example, in the latest update to its NECP, Germany includes a Soil Inventory as one of the tools to achieve its climate mitigation targets and enlists a series of measures, including both general measures—such as the review of the domestic legal basis for soil protection and amendments to soil protection legislation to better account for climate mitigation, climate adaptation, and biodiversity protection,[31] as well as more specific measures such as rewetting of peat soils, which, however, entail broader planning coordination

[24] Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions, "Securing our future: Europe's 2040 climate target and path to climate neutrality by 2050 building a sustainable, just and prosperous society" (COM/2024/63 final).

[25] Article 2(1) European Climate Law.

[26] Kulovesi et al. (2024).

[27] Article 2(2) European Climate Law.

[28] Regulation (EU) 2018/1999 of the European Parliament and of the Council of 11 December 2018 on the Governance of the Energy Union and Climate Action, amending Regulations (EC) No. 663/2009 and (EC) No. 715/2009 of the European Parliament and of the Council, Directives 94/22/EC, 98/70/EC, 2009/31/EC, 2009/73/EC, 2010/31/EU, 2012/27/EU and 2013/30/EU of the European Parliament and of the Council, Council Directives 2009/119/EC and (EU) 2015/652 and repealing Regulation (EU) No. 525/2013 of the European Parliament and of the Council.

[29] Ibid., Articles 5 and 13; Articles 6 and 7 European Climate Law.

[30] Institute for European Environmental Policy (2020).

[31] Update of the Integrated National Energy and Climate Plan – Plan of the Federal Republic of Germany, August 2024, available at: https://commission.europa.eu/document/download/cd8ba2d6-1af6-4f37-aa07-059989bb1264_en?filename=GERMANY%E2%80%93FINAL_UPDATED_NECP_2021-2030_%28ENGLISH%29.pdf, p. 121.

between the federal and sub-federal levels.[32] France, differently, includes soil's carbon sequestration as one of the policy goals, also leading existing initiatives and policies with regard to carbon farming.[33] Yet the French NECP does not delve into great detail about concrete measures to be adopted to ensure soil protection, whereas SOC stocks are mentioned with regard to financial support for sustainable farming activities.[34]

Overall and beyond pointed examples, it is accurate to state that soil protection and management are not fully integrated in the current EU Energy and Climate governance framework. The current overarching EU Energy and Climate regime in particular does not yet: a) fully account for the climate mitigation (and adaptation) benefits of soils, b) ensure that the pressures and contributions of soils are duly coordinated with other sectoral legislation with other key EU policies, foremost the renewable energy policy, and c) set specific overarching targets for soil protection and restoration—beyond those established under the EU Nature Restoration Law with regard to terrestrial biodiversity ecosystem protection.

2.2 Renewable Energy Directive (RED)

The Renewable Energy Directive (RED) sets the legal framework for renewable energy generation in the EU with the view to implementing the EU's renewable energy targets by 2030.[35] Its asserted objective is to set a common framework for the production of renewable energy and to set related binding targets.[36] The RED establishes binding EU-wide targets for renewable energy generation by 2030.[37] The 2030 target has been revised since the adoption of the European Green Deal in 2019, with a series of amendments to the RED up to 42.5%, with the aspiration to achieve 45%.[38] Granular targets are then set for Member States nationally, which are considered as 'State's minimum contribution' to the EU's Climate and Energy

[32] Id., p. 118.

[33] National Energy and Climate Plan France - update 2024, available at: https://commission.europa.eu/document/download/ab4e488b-2ae9-477f-b509-bbc194154a30_en?filename=FRANCE%20%E2%80%93%20FINAL%20UPDATED%20NECP%202021-2030%20%28English%29.pdf, p. 104.

[34] Id., pp. 95, 96, 100.

[35] Directive (EU) 2018/2001 of the European Parliament and of the Council of 11 December 2018 on the promotion of the use of energy from renewable sources.

[36] Ibid., Article 1.

[37] Ibid., Article 3.

[38] Directive (EU) 2023/2413 of the European Parliament and of the Council of 18 October 2023 amending Directive (EU) 2018/2001, Regulation (EU) 2018/1999 and Directive 98/70/EC as regards the promotion of energy from renewable sources, and repealing Council Directive (EU) 2015/652, which amended Article 3(1) of the RED.

Framework.[39] Moreover, the RED mandates Member States to require fuel suppliers to supply a minimum of 14% renewable energy in the transport sector—essentially from biofuels.[40] With regard to both the broader renewable energy generation and the transport sector targets, the approach adopted in the RED is to leave in the hands of Member States the implementation of specific policies related to the deployment of renewable energy installations and uptake of biofuels, as outlined in their NECPs submitted to the European Commission.

The EU environmental objectives have constantly been in tension with the EU energy policy objectives of ensuring security of supply and upscaling sustainable fuels and renewable energy installations while minimising impacts on ecosystems.[41] The two main instances where the tension between the EU's energy and environmental objectives arises within the RED, more specifically with regard to land and soil ecosystems conservation, are the promotion of biofuels and the deployment of large-scale renewable energy installations, including wind, solar (PV), and hydropower plants, to which we now turn.

2.2.1 Biofuels and Biomass Feedstock

Concurrently, a long-standing issue addressed by the RED is the environmental impacts of biomass and biofuels feedstock, in particular from food feedstock (such as sugar, vegetable oils, etc.).[42] The RED has also been repeatedly amended in light of ongoing concerns about the significant GHG emissions stemming from *Indirect Land Use Change* (ILUC) generated by the expansion to non-crop land of monocultures devoted to food feedstock for biofuels, foremost, palm oil, both within and outside the EU.[43] The RED sets out rules on both conventional and advanced biofuels. Conventional (or first-generation) biofuels are those produced from land using feedstocks that can also be used for food and feed. Advanced (or second or third generation) biofuels, differently, are produced from feedstock that do not compete directly with food and feed uses (such as waste and agricultural residues, non-food crops, etc.).

On the one hand, the RED sets a limit to the amount of food crop feedstock that could be taken into account by Member States towards the achievement of their national renewable energy targets in the transport sector (7%).[44] On the other hand, the RED sets out a list of *sustainability criteria* for biomass and advanced biofuels.[45] Only biomass and biofuels meeting the sustainability criteria are allowed to be

[39] Recital 10 RED.

[40] Article 25(1) RED.

[41] Montini (2024).

[42] Jeswani (2020, p. 3).

[43] Searchinger et al. (2018), European Court of Auditors (2023).

[44] Article 26(1) RED.

[45] Article 29 RED.

accounted for by Member States towards their national renewable energy targets in the transport sector.[46] The RED's sustainability criteria exclude any biofuel feedstock produced on land that has been deforested or (in the case of wetlands and peatlands) drained as of 2008.[47] In addition, all feedstock generated from agricultural land may count towards the EU renewable energy targets only when produced against a background providing for monitoring or management plans set in place to address, among others, the impacts thereof on soil quality and soil carbon.[48]

With regard to feedstock generated from forest biomass, the RED requires that Member States of provenience adopt national or sub-national plans or, in the absence thereof, that forest management practices are in place to ensure that *"harvesting is carried out considering maintenance of soil quality and biodiversity with the aim of minimizing negative impacts,"* including the prevention of harvesting on vulnerable soils, and the use of logging systems that minimise soils impacts, foremost soil compaction.[49] Lastly, additional reporting requirements are introduced for Member States and operators. Accordingly, and with specific regard to voluntary certification schemes, the European Commission could eventually mandate additional requirements for further information about soil, water, and air protection carried out by biofuels producers and suppliers as complementary to the general sustainability criteria laid out in the RED.[50]

The RED sustainability criteria have been complemented since 2019 by specific rules dedicated to the certification of low-ILUC biofuels and biomass feedstock, which apply to all feedstock suppliers to the EU.[51] The ILUC Delegated Regulation points to three key additionality criteria to define low-ILUC feedstocks, which relate to the financial viability of biofuels production, the uptake by small-sized farmers and—relevant here—the cultivation of food on abandoned or severely degraded land.[52] In this score, the ILUC Regulation aims to foster sustainable agricultural practices at a small scale and soil management with regard to the potential restorative function of biofuels feedstocks to previously degraded soils. As recently noted, *"low ILUC-risk initiatives to boost productivity on agricultural land and activity on abandoned and severely degraded land have the potential to align with policy goals for rehabilitating and regenerating soils (as well as for invigorating and diversifying rural economies)."*[53]

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[46] Article 29(1) RED.

[47] Article 29(3) RED.

[48] Article 29(2) RED.

[49] Article 29(6)(a) RED.

[50] Article 30 RED.

[51] Commission Delegated Regulation (EU) 2019/807 of 13 March 2019 supplementing Directive (EU) 2018/2001 of the European Parliament and of the Council as regards the determination of high indirect land-use change-risk feedstock for which a significant expansion of the production area into land with high carbon stock is observed and the certification of low indirect land-use change-risk biofuels, bioliquids and biomass fuels.

[52] Article 5 ILUC Delegated Regulation.

[53] Sandford et al. (2024).

Furthermore, following the 2023 amendments to the RED (REDIII), it is provided that woody biomass should be *used "according to its highest economic and environmental added value"* following the cascading principle.[54] Accordingly, *"resource efficiency of biomass use by prioritising, wherever possible, the material use of biomass over its energy use, thus increasing the amount of biomass available within the system."*[55] The cascading principle in the REDIII introduces a structured hierarchy for the use of wood, similar to the waste hierarchy. It prioritises the use of wood in long-lived products, such as furniture and building materials, before considering options for repair, reuse, or recycling. Woody biomass should be used according to its highest economic and environmental added value in the following order of priorities: wood-based products, extending the service life of wood-based products, re-use, recycling, bioenergy, and disposal. The cascading principle could, however, be derogated in light of energy security concerns or where there are no industries or processing facilities that could make higher added value use of certain feedstocks within a geographical perimeter.[56] To date, the only policy improvement compared to RED II is the ban on biomass use from primary forests, peatlands and wetlands.[57] However, this ban also requires implementation, monitoring, control and enforcement, which is not guaranteed. Carbon-rich forest types such as old-growth forests might, in contrast, still be used for energy production without further restrictions beyond the slight update of the sustainability criteria.

2.2.2 Renewable Energy Installations Permitting and Planning

With regard to large-scale renewable energy installations, very little guidance is given in the RED as to how to deal with soil sealing and any land ecosystem impacts generated therefrom. The assumption here is that the ultimate decisions regarding soil and land impacts stemming from large-scale installations would be addressed by Member States through land planning and point impact assessments. This picture, however, drastically changed after 2022 in light of the energy crisis caused by the outbreak of the war in Ukraine. The EU's policy response to the unfolding crisis was the REPowerEU plan.[58] The REPowerEU plan aims chiefly to diversify the supply of natural gas in order to reduce dependency on Russia and to foster the uptake of renewable energy generation, with particular regard to large-scale wind and solar PV installations.[59] Renewable energy facilities are thus seen as key infrastructures

[54] Recital 10 REDIII; Article 3(3) RED.

[55] Ibid.

[56] Article 3(3a) RED.

[57] Article 29(6) RED.

[58] Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions RePowerEU Plan (COM/2022/230 final).

[59] Id., pp. 6–7.

to ensure energy security while achieving the European Climate Law and the RED targets.[60]

Against this backdrop, two of the key problems identified by the European Commission for the uptake of renewable energy infrastructures are spatial planning and permit-granting at the Member States' level.[61] The so-called 'Emergency Regulation' was hence adopted in late 2022, *"to introduce additional urgent and targeted measures addressed to specific technologies and types of projects which have the highest potential for quick deployment and immediate effect on the objectives of reducing price volatility and reducing the demand for natural gas without constraining the overall energy demand"* for a period of 18 months.[62] The Emergency Regulation was adopted under the legal basis of Article 122 TFEU, which enables the Council to adopt legislation in the field of energy in emergency situations. Yet importantly, these emergency provisions have been subsequently crystallised in the REDIII introduced in 2023, as well as in the legislation regarding energy projects of common interest under the revised Trans European Network (TEN-E) Regulation.[63]

The REDIII now establishes a fast-track procedure to streamline permit-granting and land planning for renewable energy installations and related equipment and infrastructure. The fast-track procedure revolves around a general rebuttable presumption that the planning, construction and operation of renewable energy installations and related infrastructures (including transmission networks and storage assets) are deemed as being of *"overriding public interest and serving public health and safety"* for the purposes of the relevant EU environmental legislation.[64] This clause operates with regard to the impact assessments required pursuant to the

[60] Ibid.

[61] Recital 20 of Directive (EU) 2023/2413 of the European Parliament and of the Council of 18 October 2023 amending Directive (EU) 2018/2001, Regulation (EU) 2018/1999 and Directive 98/70/EC as regards the promotion of energy from renewable sources, and repealing Council Directive (EU) 2015/652. See also the explanatory memorandum to the proposal for the same Directive, according to which: *"For some wide-spread renewable technologies, such as wind power and photovoltaics, administrative barriers are increasingly becoming more important than other types of barriers. (…) The most common barriers related to the administrative process for renewable energy (…) Conflicting public goods are the second main source of obstacles for the deployment of renewable installations. (…) The most prominent among them concern environmental protection (biodiversity and protection of endangered species and protection of water bodies)."*

[62] Council Regulation (EU) 2022/2577 of 22 December 2022 laying down a framework to accelerate the deployment of renewable energy, Recital 5.

[63] Regulation (EU) 2022/869 of the European Parliament and of the Council of 30 May 2022 on guidelines for trans-European energy infrastructure, amending Regulations (EC) No. 715/2009, (EU) 2019/942 and (EU) 2019/943 and Directives 2009/73/EC and (EU) 2019/944, and repealing Regulation (EU) No. 347/2013.

[64] Article 3 Emergency Regulation; Article 16f RED.

Habitats Directive,[65] the Water Framework Directive[66] and the Birds Directive[67] (2009/147/EC).[68] This implies that Member States shall give priority to renewable energy infrastructure projects when balancing interests in any given case, except for instances *where "there is clear evidence that those projects have significant adverse effects on the environment which cannot be mitigated or compensated for."*[69] The only backstop to the presumption of overriding public interest relates to the adoption of appropriate species conservation measures or related financial resources, as well as the availability of areas for restoration or conservation of species eventually affected, whereas no provisions are made for the protection of soil and land ecosystems.[70] The presumption of overriding public interest applies, in the words of the REDIII, *"until climate neutrality is achieved."*[71]

From the land-planning perspective, under the REDIII, by February 2026, Member States shall map and designate *'renewable acceleration areas'*.[72] The *'renewable acceleration areas'* are areas deemed necessary for each Member State to achieve its national contribution towards the EU's 2030 overall renewable energy target.[73] In the words of the European Commission: *"Member States should swiftly identify suitable land and sea areas for renewable energy projects, commensurate with their national energy and climate plans and their contribution to reaching the revised 2030 renewable energy target. As part of this mapping process, limited and clearly defined areas should be designated as particularly suitable for the development of renewable energy (renewable go-to areas), while avoiding as much as possible environmentally valuable areas and prioritising inter alia degraded land not usable for agriculture."*[74]

At the same time, Member States should limit 'exclusion zones' where renewable energy cannot be developed to the necessary minimum.[75] The main criteria to identify renewable acceleration areas relate to the potential for renewable energy generation on the land surface, the projected demand for energy, and the availability of energy infrastructures, including grids and storage.[76] Member States are therefore

[65] Council Directive 92/43/EEC of 21 May 1992 on the conservation of natural habitats and of wild fauna and flora.

[66] Directive 2000/60/EC of the European Parliament and of the Council of 23 October 2000 establishing a framework for Community action in the field of water policy.

[67] Directive 2009/147/EC of the European Parliament and of the Council of 30 November 2009 on the conservation of wild birds.

[68] Article 16f RED.

[69] Article 3(2) Emergency Regulation; Recital 44 Directive 2023/2413/EU amending the RED.

[70] Article 16a(5) RED.

[71] Article 16f RED.

[72] Article 15b and Article 15c RED.

[73] Recital 26 Directive 2023/2413/EU.

[74] EC Recommendations on speeding-up permit granting for renewable energy infrastructures, C/2022/3219 final, para 21.

[75] Id., para 22.

[76] Article 15b RED.

required to develop plans to implement renewable energy projects in the designated renewable acceleration areas.[77] In doing so, Member States should identify areas where renewable energy projects would not have significant environmental impacts based on sensitivity mapping. It is important to note that no specific provisions are laid out for soil and land integrity, and the focus lies primarily on species and habitat conservation. Nonetheless, such plans will be subject to Strategic Environmental Assessment, which arguably should account for potential impacts on soil ecosystems, although not on a project-specific basis. Yet all the projects falling in the renewable acceleration areas shall be exempted from any requirements under the EIA Directive and EU biodiversity legislation.

Three main conclusions can be drawn from the analysis carried out in this section. First, soil management and protection are not fully integrated into the EU's Climate and Energy framework. Second, the RED sets out general requirements for the protection of soils and the recovery of degraded land in relation to the production of biofuels feedstocks and biomass, albeit mostly indirectly. Third, under the RED soils, as well as broader environmental impacts, are significantly sidelined in both planning and permitting requirements regarding the deployment of renewable energy installations.

3 The Broader Regulatory Context of the Energy Transition

Other than instruments directly addressing the renewable energy sector, a broader view should be taken on the regulatory context needed for achieving carbon neutrality. We will consider if and how critical raw materials and sustainable investment laws—necessary conditions for the energy transition—address soil health.

3.1 Critical Raw Materials

The challenges of access to critical raw materials have been the subject of several policy acts,[78] culminating in the adoption of the Critical Raw Materials Act[79] in 2024. The latter aimed at ensuring a secure and sustainable supply. Critical raw

[77] Article 16b RED.

[78] The Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions, Critical Raw Materials Resilience: Charting a Path towards greater Security and Sustainability COM (2020) 474 final.

[79] Regulation (EU) 2024/1252 of the European Parliament and of the Council of 11 April 2024 establishing a framework for ensuring a secure and sustainable supply of critical raw materials and amending Regulations (EU) No. 168/2013, (EU) 2018/858, (EU) 2018/1724 and (EU) 2019/1020.

materials are indispensable for the energy transition,[80] as the supply chains required for the technologies of two key sectors—renewable energy and e-mobility—depend on many of them.[81] Significant dependency on third countries and the resulting risks to the European economy and autonomy are key reasons for adopting these acts. As a consequence, this Act establishes goals ('benchmarks') for the EU's extraction, processing, and recycling capacities in 2030, in line with the energy targets set out in the European Climate Law and the Digital Decade Policy Programme.[82] We can therefore conclude that the energy transition requires an increase in the EU's extraction and processing capacities, which notably present risks to soil health.

These risks are not ignored by the EU. The Critical Raw Materials Act assumes the potential of impacts on soil health and refers to environmental assessments and authorisations already required by EU law as adequate ways to minimise and prevent them. However, special procedural acceleration tools are allowed for strategic projects. These raw material projects, required to have the potential for a meaningful contribution to EU goals in a reasonable timeframe,[83] are subject to special reporting and information duties and benefit from a special status. To derogate rules enshrined in the Birds Directive, strategic projects are automatically considered to be of public interest or serving public health and safety and may be considered as having overriding public interest. The same status is relevant for exempting a Member State from being considered in breach of the Water Framework Directive and to overcome a negative assessment of the project under the Habitats Directive. As a consequence, the special status granted to strategic projects significantly reduces the strength of the goal of maintaining soil ecological integrity before the goals of energy transition. Moreover, strategic projects are also granted the highest priority status, and several norms ensure the acceleration of administrative procedures and court decisions.[84] This is in line with the similar norms in the RED, addressed in Sect. 2.2.2.

A sustainability chapter is included in the Critical Raw Materials Act, dealing with circularity measures. The national measures are to be designed to ensure circularity[85] are in line with the tools already foreseen in the Circular Economy Action Plan,[86] aiming, among other objectives, to increase resource efficiency and promote the use of secondary critical raw materials, thereby reducing the need for more extraction. If effective, these measures are well-suited to reducing extraction both within the EU and abroad, with a clear positive impact on soil ecological integrity.

[80] Pommeret et al. (2022). This is also recognised in the Critical Raw Materials Act (cf. Recital 2).

[81] European Commission (2020).

[82] Critical Raw Materials Act, Article 5 and Recital 11.

[83] Criteria are defined in Article 6.

[84] Cf. Articles 10 (3–5), 11, 12, and 15 Critical Raw Materials Act.

[85] Articles 26 to 29 Critical Raw Materials Act, namely by incentivising re-use, repair, collection, and recycling.

[86] Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. *"A new Circular Economy Action Plan For a cleaner and more competitive Europe."* COM/2020/98 final.

By minimising the excavation of large amounts of rocks to extract from ores and geological deposits, they have the potential to help prevent the disruption of soil ecosystems and the contamination of soil and water.

3.2 Sustainable Investment

EU climate and energy targets also depend on investment, including in what concerns energy sector.[87] The opportunities presented by renewable energies transpire from reports concerning the investment outlook in the sector.[88] This means that sustainability standards concerning both financial and non-financial sectors assume a very significant role in shaping the real environmental impact of the energy transition, including regarding soil. This is relevant not only for soil protection in the EU, but also, crucially, abroad. The Corporate Sustainability Due Diligence Directive expressly acknowledges the particular relevance of critical raw materials in global value chains and their tremendous potential environmental impact.[89] As a consequence, the criteria for defining what can be claimed as sustainable in investments in energy projects must be scrutinised to identify how the impacts on the ecological integrity of the soil have been taken into account, or if unsustainable soil management practices are negative requirements under the 'no significant harm' condition.

The Taxonomy Regulation presents a common framework for determining whether an economic activity, such as the production of energy, or upstream activities, such as raw materials extraction, qualifies as environmentally sustainable. It should be highlighted that no explicit or specific mention is made of soil in the list of environmental objectives that should not be significantly harmed by these activities.[90] In this regard, and considering only what is directly relevant in terms of energy projects and soil protection, the Taxonomy Regulation recognises 'sustainable land use and management' as a means to making a substantial contribution to the protection and restoration of biodiversity and ecosystems in Article 15 (1)(b), with a positive impact in mitigating climate change. This may include protection of soil biodiversity, land degradation neutrality and remediation of contaminated sites.

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[87] As assumed in the Action Plan: Financing Sustainable Growth, to address the need to reorient capital flows, considering that *"Europe has to close a yearly investment gap of almost EUR 180 billion to achieve EU climate and energy targets by 2030"* – cf. Communication from the Commission to the European Parliament, the European Council, the Council, the European Central Bank, the European Economic and Social Committee and the Committee of the Regions, "Action Plan: Financing Sustainable Growth" (COM/2018/97 final), p. 2.

[88] KPMG (2024).

[89] Directive (EU) 2024/1760 of the European Parliament and of the Council of 13 June 2024 on corporate sustainability due diligence and amending Directive (EU) 2019/1937 and Regulation (EU) 2023/2859, recitals 9, 25, 68 and Article 3 (1g).

[90] Cfr. Articles 9 and 17 Taxonomy Regulation.

The Taxonomy Regulation prescribes the approval of several delegated acts for determining the fulfilment of its several conditions. Commission Delegated Regulation (EU) 2021/2139 details the screening criteria for determining the conditions under which an activity qualifies as contributing substantially to climate change mitigation or adaptation and whether that activity causes no significant harm to any of the other environmental objectives. We will now provide an overview of how impacts on soil, either positive or negative, were considered in activities relevant to the energy transition.

In what concerns activities with significant impact on soil health, the positive impacts of forest and waste management activities for the production of renewable energy are notably argued in detail in several recitals (e.g., 23, 25, 32). However, negative impacts are sparingly considered. Let us use the example of Forestry activities (Annex I, Section 1), arguably a very relevant sector for renewable energy production. The 'do no significant harm' conditions are aggregated into sections concerning factors such as sustainable use and protection of water and marine resources, pollution prevention and control, and protection and restoration of biodiversity and ecosystems. The section regarding water resources refers to existing EU law acts, such as water protection, chemicals and environmental impact assessment rules[91] and provides a generic mention that *"pollution of water and soil is prevented and cleaning up measures are undertaken when pollution occurs."*[92] In what refers to biodiversity and ecosystems, information must be provided on *"ensuring the maintenance and improvement of physical, chemical and biological quality of the soil,"* but no specific negative conditions are determined based on this information.[93] The forest management and conservation forestry sections add that sustainable activities must not involve the *'degradation of land with high carbon stock'* and that measures are taken to avoid the use of certain hazardous chemicals and pesticides.[94] In what concerns some activities in the energy sector, the amendments introduced by Commission Delegated Regulation (EU) 2022/1214 added a technical screening criterion that radioactive discharges to soil should comply with individual license conditions.[95]

[91] Namely by numerous referrals in Annex I to Appendixes B, C, and D Commission Delegated Regulation (EU) 2021/2139.

[92] Annex I, 1. Forestry, 1.1. Afforestation, Do no significant harm ('DNSH'), (5) Pollution prevention and control. The same provisions can be found in the equivalent Section of 1.2. Rehabilitation and restoration of forests, including reforestation and natural forest regeneration after an extreme event Commission Delegated Regulation (EU) 2021/2139.

[93] Annex I, 1. Forestry, 1.1. Afforestation, Do no significant harm ('DNSH'), (6) Protection and restoration of biodiversity and ecosystems, d), Commission Delegated Regulation (EU) 2021/2139. The same provisions can be found in the equivalent Section of 1.2. Rehabilitation and restoration of forests, including reforestation and natural forest regeneration after an extreme event.

[94] Annex I, 1. Forestry, 1.3. Forest management, Do no significant harm ('DNSH'), (1) Climate change mitigation, 1.3.c). The same provisions can be found in the equivalent Section of 1.4. Conservation forestry, Commission Delegated Regulation (EU) 2021/2139.

[95] Annex I introduced by the Commission Delegated Regulation (EU) 2021/2139.

Indirect soil management measures are also sparingly included in the remaining sections of the Delegated Regulation, particularly in the one dedicated to *'environmental protection and restoration activities'*, which includes only one type of activity: restoration of wetlands. The section dedicated to energy is entirely devoid of any consideration of negative impacts on soil ecological integrity. As a side note—since it does not reflect directly on the energy transition—it is also noteworthy that protection of arable land and crop land is safeguarded in the section concerning construction of buildings; in contrast, only greenfield land of recognised high biodiversity value and land that serves as habitats to endangered species has enough relevance to exclude construction as being considered sustainable.[96] This seems to evidence an overall low standard of protection for natural elements, but particularly so regarding soil and its ecological functions.

We can therefore conclude that the screening criteria for determining whether an activity relevant for energy transition can be considered as a sustainable investment only laterally or indirectly consider soil ecological integrity, in line with what results from the analysis of other EU legal acts. Several actions have been brought before the Court of Justice of the European Union against the taxonomy regulation framework, both by states and non-governmental organisations.[97] These cases present several claims, both procedural and substantive; none of them, however, seems to address soil protection or integrity.

4 The Other Side of the Mirror: Energy Transition in Nature Restoration Law

We have so far analysed how soil integrity and sustainable management have been considered in policies and laws designed to enhance energy transition goals, as would be expected under the command that environmental concerns must be integrated into all sectoral policies. We have concluded that, overall, the EU legal framework on climate and energy sparingly considers soil protection and expressly decreases the relevance of ecological integrity by consistently recognising energy projects as having overridden public interest in environmental authorisation and assessment procedures.

We will now review how energy transition policies have been addressed in a law directly aiming to protect ecological values. Restoring ecosystems by broadening

[96] Annex I, 7. Construction and real estate, 7.1. Construction of new buildings, Do no significant harm ('DNSH'), (6) Protection and restoration of biodiversity and ecosystems, Commission Delegated Regulation (EU) 2021/2139.

[97] *ClientEarth v Commission*, Case T-579/22 (pending); *ClientEarth and Others v Commission*, Case T-215-23 (pending); *Greenpeace and Others v Commission*, Case T-214/23 (pending); *Robin Wood and Others v Commission*, Case T-575/22 (pending); *Fédération environnement durable and Others v Commission*, Case T-583/22 (pending); *Dryade, Fossielvrij NL, and Protect our Winters Austria and Others v EU Commission*, Case T-449-24 (pending).

nature conservation beyond existing protected areas is the declared goal of the Nature Restoration Law.[98] The relevance of this act for protecting soil, as an integral part of ecosystems, is obvious. In fact, the recitals establish a clear link with the EU Soil strategy for 2030[99] for which the Nature Restoration Law is a crucial piece, since a specific legal framework for soil protection has so far failed to be adopted.

The primary obligation for Member States is to adopt restoration measures.[100] Specific duties are imposed on Member States, with goals and timelines depending on the type of habitat.[101] For this chapter, we will consider the duty to adopt measures to ensure continuous improvement and/or prevent significant deterioration in the condition of the habitat types. These obligations, however, will not apply to a plan or project of (i) overriding public interest when (ii) there are no less damaging alternatives. In what concerns the first of these conditions, all "*planning, construction and operation of plants for the production of energy from renewable sources, their connection to the grid and the related grid itself, and storage assets"* are presumed to be of overriding public interest.[102] Moreover, it is at the Member States' discretion to dismiss the second condition when the plan was subject to a strategic environmental impact assessment or when the project went through an environmental impact assessment.[103] It should be noted that no substantive conditions are imposed in what concerns the result of these assessments.

In implementing these obligations, Member States are required to adopt national restoration plans. These plans must consider the renewable energy targets, mapping areas that are required to fulfil them, including the designation of the renewable energy acceleration areas and dedicated infrastructure areas.[104] We can therefore conclude, also in light of the above analysis of the RED, that energy projects' impacts are practically exempt from nature restoration obligations and that the primacy of energy policies results not only from energy law and policies, but also from nature restoration laws.

[98] Regulation (EU) 2024/1991 of the European Parliament and of the Council of 24 June 2024 on nature restoration and amending Regulation (EU) 2022/869.

[99] Recitals 23, 57.

[100] Articles 4 and 5.

[101] Cf. de Leeuw and Backes (2024), Hardiman (2022), Lees and Pedersen (2025).

[102] Article 6(1).

[103] Member States can restrict this to (i) certain parts of their territory, (ii) certain types of technologies, or (iii) to projects with certain technical characteristics in accordance with the priorities set in their integrated national energy and climate plans. Cf. Article 6 (2).

[104] Articles 14 (13) and 15(3).

5 Discussion

The general requirements under the EU Climate Law and the EU Governance Regulation, based on the broader framework of the EU Energy Union, chiefly address GHG emissions reductions and offsetting. The planning and reporting requirements established for Member States thereunder in the NECPs do not encompass specific elements about the protection of sensitive soils against competing uses, especially for energy generation. This leaves ample room for Member States to deal with soil management and protection, thus ultimately leaving up such embeddedness to different domestic pushes for environmental innovation.[105]

The RED, as a flagship legislation of the EU's renewable energy policy, deals with the soil vs. energy conundrum in two respects. First, it sets out a set of standards of biofuels and biomass feedstock with a view to creating a governance effect to that end.[106] The RED seeks to promote low ILUC-risk non-food feedstocks through sustainability criteria and by applying cascading principles for the use of wood biomass to minimise impacts on land and soil ecosystems. Initiatives to boost productivity on agricultural land and activity on abandoned and severely degraded land have the potential to align with policy goals for rehabilitating and regenerating soils (as well as for invigorating and diversifying rural economies).[107] Yet low ILUC-risk categorisation does not address the conflict between using land for agriculture and protecting natural areas, or between using agricultural inputs to increase yields and minimise environmental pollution while conserving resources like water and soil carbon. The RED explicitly affords Member States a degree of autonomy in their treatment of biofuels' ILUC impacts, which can be exploited in the implementation of the RED to ensure that biofuels-related projects take due account of soil impacts.[108] Enforcement and control issues associated with difficult monitoring and general depiction issues persist, which would require stronger implementation support, such as enough funding and enforcement. Moreover, at least for EU-based projects, there is scope for more explicit co-recognition of sustainable agricultural practices that meet Common Agricultural Policy (CAP) goals; this would, however, require a considerable level of coordination within the European Commission. In this regard, the European Commission's proposal for a Soil Monitoring and Resilience Law includes crop rotations and cover cropping among sustainable soil management principles, as well as the use of deep-rooting perennials, which might allow scope for integrating low-input bioenergy production.[109]

Second, the RED addresses the deployment of large-scale renewable energy projects, including both installations and related infrastructures. As of 2022, the

[105] Turpin et al. (2017, pp. 241–249).

[106] Stubenrauch and Garske (2023).

[107] Paul (2023).

[108] Article 28(1) RED.

[109] Proposal for a Directive of the European Parliament and of the Council on Soil Monitoring and Resilience (Soil Monitoring Law) (COM/2023/416 final), Annex III.

EU's energy policy has clearly shifted its focus towards energy security.[110] This shift is seriously at odds with the original commitment endorsed by the EU under the European Green Deal in 2019 to ensure ecosystems' integrity and, to a significant extent, the stated goals of ensuring soil health, soil protection, and restoration as stated in the Soil Strategy for 2030 and the EU Biodiversity Strategy for 2030.[111] A clear manifestation of this interplay is displayed by the new legal framework for renewable energy infrastructure planning and permit-granting adopted in the REDIII.[112] Accordingly, substantial bypassing of impact assessment requirements under EU Environmental legislation is allowed for a large share of local and cross-border strategic renewable energy projects, including installation and ancillary infrastructures (grid, transmission, storage, etc.). The general counterbalancing element envisaged in the REDIII is the adoption or provision of mitigation and compensatory measures by project developers. Our review, therefore, concludes that soil protection and management are not fully integrated into the EU's Climate and Energy law and policy.

The same conclusion arises from an analysis of acts that, while not directly regulating energy production, establish fundamental conditions for it. The Critical Raw Materials Act, which addresses materials essential for renewable energy production and storage, creates a special status for strategic projects, allowing them to benefit from special rules, including procedural accelerations. More importantly, these projects are granted a public interest status, creating an open gateway to what should be exceptions under the Birds Directive, the Water Framework Directive, and the Habitats Directive. Considering that soil protection in EU law is mostly indirect and achieved through these legal acts, we can conclude that the priority given to these projects weakens the goal of maintaining soil ecological integrity in favour of advancing the energy transition.

This is in sharp contrast with what had been the high standard required by the Court of Justice of the European Union, when stating that the assessment of compensatory or mitigation measures by national authorities should be carried out narrowly and should not, by any means, affect the integrity of the environment in light of the precautionary principle.[113] These broad exemptions resulting from both RED and the Critical Raw Materials Acts represent, therefore, a significant decrease in the level of environmental protection when dealing with energy projects. This also follows from a review of Nature Restoration law, which largely exempts energy projects from restoration obligations. As a result, the imbalance in favour of energy goals and present in energy law is not offset by higher protection standards in environmental laws.

[110] Shaar and Leal-Arcas (2024).

[111] Jendrośka and Anapyanova (2023).

[112] Montini (2024).

[113] *Briels v Minister van Infrastructuur en Milieu*, Judgment of the Court (Second Chamber), 15 May 2014, Case C-521/12.

A second material condition for energy transition is a significant flow of investment, which can benefit from classification as a sustainable investment due to its contribution to climate change mitigation. A careful look at the screening criteria adopted by the European Commission for determining whether an investment is sustainable reveals an almost complete lack of consideration of negative impacts on soil. This conclusion arises from the delegated acts but is also a clear consequence of the absence of any specific mention of soil protection in the environmental objectives set out as negative conditions under the 'no significant harm' principle. This might be justified, once again, by the lack of a legal framework for soil health. However, it reveals a striking discrepancy in the standards of ecological protection, particularly when compared to water and biodiversity protection.

A dim positive outlook might be found in the investment in circularity measures, which might, overall, decrease the need for more energy and extraction. A second potentially positive prospect is the ongoing proposal for a Soil Monitoring and Resilience Law. With a very different political and legal backdrop, the 2006 proposal unsurprisingly included little reference to energy policies. However, a very relevant provision enshrined a duty to identify, describe, and assess the impacts of energy policies on soil.[114] The ongoing proposal mentions renewable energy for the purpose of ensuring energy policy goals and highlighting the synergies created by soil health and energy transition.[115] However, it also emphasises the need for Member States to integrate sustainable soil management practices in an array of plans and programmes, importantly including the NECPs.[116] Soil management practices could then be streamlined across national energy planning, leading to concrete measures to regenerate soil ecosystem functions, including carbon sequestration, while overall minimising physical soil disturbance.[117] This would be a promising development, all the more in light of the ongoing developments under EU renewable energy legislation.

6 Conclusion: Soil as a Missing Piece in the EU's Climate, Energy and Environment Conundrum

An analysis of EU legislation dealing with the core interplay between soil and land protection and energy generation further unfolds the existing policy 'schism' within the European Union between climate change, energy, and environmental protection.[118] Soil protection and management are insufficiently integrated into the EU's Climate and Energy policy framework. While EU climate legislation prioritises

[114] Article 3.

[115] Recital 28.

[116] Article 10 and Annex IV.

[117] Hannam (2024b), Fermeglia (2022).

[118] Zahar (2020).

GHG emissions reductions, it largely overlooks the specific needs of soil ecosystems, especially in contexts where land is under pressure from energy transition infrastructure and renewable energy projects. This also becomes clear when analysing RED's attempts to address the soil-energy conflict, particularly when considering how large-scale renewable energy projects have been facilitated by recent legal reforms that allow for exemptions from environmental assessments, prioritising energy development over ecosystem protection.

This imbalance is further reinforced by the Critical Raw Materials Act, which gives strategic energy-related projects a special legal status, weakening the application of general environmental law. The Nature Restoration Law does little to offset these effects, as it also exempts energy projects from restoration obligations, further deepening the disconnect between environmental and energy objectives. In the realm of sustainable investment, the EU's taxonomy framework also shows a near-total neglect of soil impacts, with barely any explicit recognition of soil in the 'do no significant harm' criteria.

These inconsistencies are against the general objectives of the European Union, which is committed to achieving a high level of protection and improvement of the environment under Articles 3(3) TEU, 191(1) TFEU and 37 of the Charter of Fundamental Rights of the EU.[119] In addition, environmental protection considerations must be integrated into every other policy silos under Article 11 TFEU[120]—also to attain sustainable development[121]—and more specifically in the realm of the EU's energy policy under Article 194(1) TFEU. Moreover, the lack of integration of climate and environmental policies is but one of the manifest shortcomings of the European Green Deal in terms of both its overall ambition,[122] as well as the mainstreaming of environmental protection across other EU policy domains, thereunder.[123] Yet this is utterly problematic as legal responses to soil protection should be able to *"adapt(…) to the problems that arise from the changing characteristics of the climate and the impacts on the soil environment."*[124]

This shattered landscape also reflects a broader systemic gap, stemming from the absence of a dedicated legal framework for soil health in EU law. The proposal for a Soil Monitoring and Resilience Law offers a promising development. It proposes integrating sustainable soil management into national energy and climate plans, potentially aligning energy transition policies with soil regeneration and climate adaptation goals. At the moment, however, the hard choice between soil ecosystems' protection and the need to adopt land-ravenous technologies such as renewable energy installations appears to be solved in EU law in favour of the latter, by putting security of supply and energy independence as a priority over ecosystems' integrity, including soil security.

[119] Misonne (2015), Scotford (2018).

[120] McIntyre (2013), Karageorgou (2023).

[121] Sjåfjell (2014).

[122] Montanarella and Panagos (2021), Heuser and Itey (2022), Heuser (2022).

[123] Montini (2021)

[124] Hannam (2024a, p. 57).

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